

• **Context and Problem Statement:**

The core problem of the SLFs is the fact that they are landing on grape vines and infesting grapes. Once the grapes get infested, there are no current effective methods to separate SLFs from harvested grapes. Our objective is to prevent the issue before it becomes one, to prevent the SLFs from reaching the vines in the first place. To tackle this, we opted for a fly trap that uses light as a lure. This allowed us to avoid chemicals such as pesticides that are harmful to grapes. The idea was to place these traps outside of grapefields at night to attract and trap SLFs, preventing them from getting to the grapes.

• **Final Prototype and Application:** briefly explain what you built, how it works at the level relevant to the client, and how it will be used.

The device we have fabricated is a light trap that lures SLF away from the grape vines and traps them until they can be disposed of. The main mechanism of the trap is a clear, spinning plate with ridges that allow light to pass through, attracting SLF. The device will work by allowing SLFs to land on the plate and forcing them to either move into the tube at the center of the trap or be crushed by the interlocking ridges. The trapped bugs will be stored in the bottom container of our device and can be emptied at the user's convenience. The user can then either throw away and clean out the SLF from the container or reuse them as feed for animals.

Our next steps for its use in vineyards include slight modifications and placement strategies. This would include a version of our device where a motor would be used in place of a crank for ease and efficiency of use. We would also advise the users to place the device in areas where SLF tend to often land, therefore attracting them towards the trap rather than the grape vines. This would allow the user to achieve a more successful harvest by trapping and luring away a higher percentage of SLF from the grapevines with our device, resulting in better profit margins due to less wasted profits. Each device would require a small power device due to the addition of the motor and the existing light source. This power source could be solar powered due to its inactive nature during the daytime and the fact that it would be kept outside at all times. This device would also need to be periodically cleaned due to the chance that some SLF may get squished by our device instead of being trapped.

Assembly Process:

Only two batches of assembly need to take place to build our device. First the crank assembly needs to be put together, then the top assembly, and then the device is fully assembled.

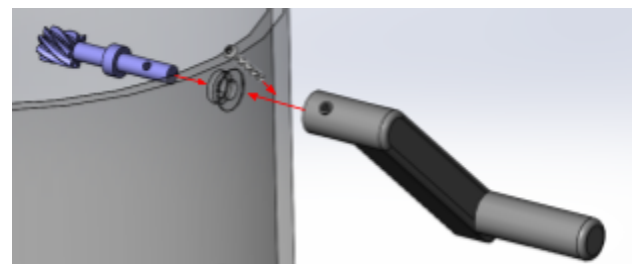


Fig #: Crank mechanism is assembled by pushing the helical pinion through the wall of the device and into the crank. A bolt then goes through both to secure.

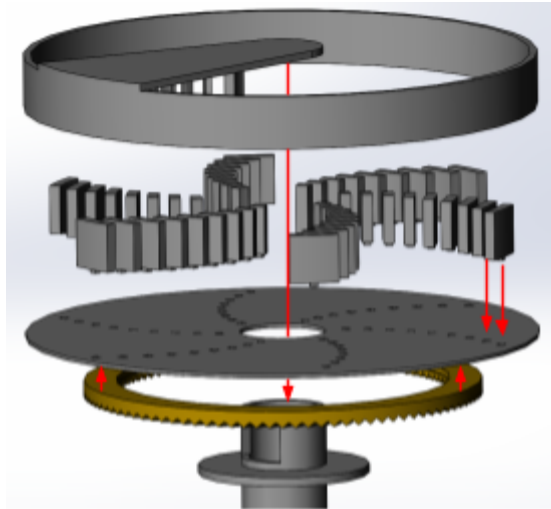


Fig #: The top part of the device is assembled by gluing the crown gear to the acrylic plate, then gluing the ridges into their holes, and then locking everything in by gluing the roof to the tube.

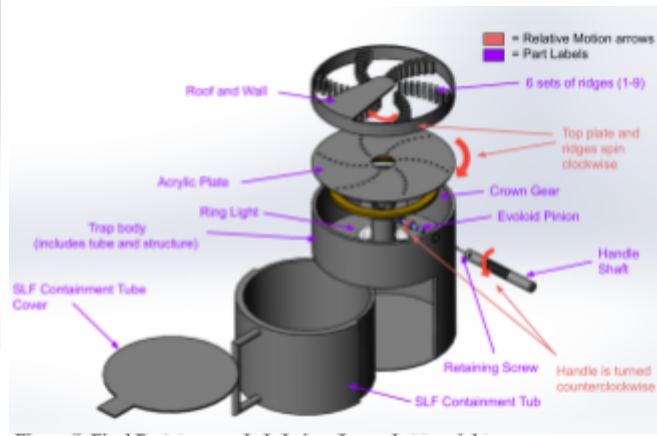


Figure #: Final Prototype exploded view. Legend at top right.

• **Conclusion and Recommendation:** state clearly what you recommend as the next step.

Is this promising and worth further development, in need of a substantial redesign, ready for field-testing? Back up your recommendation with your test results.

- We need to mention that there isn't substantial research on what kind of light attracts SLF
- Could not find big enough crown gear on McMaster so we had to 3D print it, which means it will wear down, ideally we could buy one

For the next steps of our device, we recommend both testing and slight alterations. Regarding alternation, for real-world applications, the hand crank would need to be replaced with a motor, and a solar-powered energy source would need to be explored to avoid limiting the device's placement to areas near outlets. Secondly, because the gears in our prototype were 3D printed due to McMaster not having a large enough crown gear, they would wear down significantly over time with a motor. It would be necessary to find a crown gear and a pinion to purchase, or, less ideally, a company willing to manufacture them. There would also need to be real-world testing to identify the type of light that will best attract the SLF at night. Additionally, testing our mechanism on real SLF and not our modeled ones, we tested with will be crucial for identifying any weaknesses. Overall, our device has a relatively low cost of raw materials of about \$106 and has minimal assembly, as well as easily fabricable parts that would make further exploration logical.

Testing and Results:

In order to ensure that our prototype met our success criteria, we designed tests that would provide an accurate measure of its effectiveness. Firstly, we wanted our prototype to have a large capacity for SLF so it would not have to be serviced as often. We set a target capacity of 1000 SLF as a large overestimate of

how many SLF would enter the trap per cycle. Indeed, according to a [report by Heather and Ashley Leach](#), an average of around 50 SLF were observed entering the target vineyards per day during peak season, so our device would be able to go at least a week without service. We tested this by ensuring that the bottom drawer would open and close consistently and by ensuring that its volume was great enough to hold the SLF. We estimated that we would need a minimum of 4L of storage space to hold 1000 SLF, and we checked in CAD to ensure that we achieved this volume. Given that this is a smaller model of what the final product could be, a larger version could potentially go multiple weeks without needing cleaning, making it easier for grape farmers to manage.

The second success criteria we established was that our trap should emit and disperse light effectively. Specifically, we wanted at least 80% of the light from our light source to pass through the trap top in order to minimize power demands and ensure that the strong light didn't end up overheating the trap. We tested this by measuring the lux coming from our ring light both inside and outside of our trap with a phone app that utilized the phone light sensor. We held the phone 5 in above the light and used a piece of paper as a diffuser. We found that the light dropped very little – from 14400 to 14200 lux – and determined that 98.6% of the light was able to pass through the trap. Our design will be able to let through sufficient light to lure the SLF without expending massive amounts of power.



Figure #: Testing light emission

Our third success criteria was that our trap could handle multiple SLF landing on a section of it at once, as could occur in actual use. We did not have sufficient data on how SLF are attracted to light, we set a target of handling 3 SLF landing on the trap at once, which we estimated was a reasonable figure. We made small SLF models out of paper towel and tape and attempted to push them through the trap. On our first prototype, where it was easier to simulate the motion of the SLF by pushing them, we were able to fit 6 SLF models through the central tube at once without any getting stuck. On our final, we were able to shake the trap to simulate SLF crawling through the central tube, and 3 SLF were able to be shaken into the holding container at the bottom. This indicates that our trap will be able to effectively handle trapping multiple flies at once.

Our final success criteria was that the trap not allow SLF to escape once they were on the plate, forcing them to either be crushed, or to travel down the central tube. We tested this by placing our SLF models on the top and rotating it to see if the flies would be able to escape. When the paper towel flies got trapped between the upper and lower ridges, there were no gaps for them to crawl through, and we were unable to spin the trap any more, implying that when constructed with stronger materials and faced with the weaker structure of the bugs, the threat of death would successfully be established. This ensures that the SLF attracted to our trap will be unable to escape, and will be forced to travel into the trap's center or perish.

We also ran some basic structural tests to simulate real world conditions such as inclement weather. We shook the machine at 5 Hz, spun the top at 7.25 rad/s, and threw our mock SLF at it to simulate flies landing on the top. It held up through all of those tests, implying that it will not need service often.

• Prototype and Testing Details:

Leading up to our final prototype, we built an initial prototype to confirm the functionality of our main mechanism, the spinning top. The main functionality we were hoping to confirm was that the ridges and roof ridges didn't collide, and that a threat of death was established by the ridges which would force them down the tube.

This prototype consisted of the upper components of our final device – the tube, acrylic plate, ridges and roof. A string could then be wrapped around screws placed at the edge of the plate, and then pulled, to spin the device. By testing the spinning functionality of the acrylic plate, we were confident that when adding more sets of ridges on the final prototype, everything would work as intended. Additionally, we tested to make sure that the threat of death was established by the ridges, while also forcing SLFs down the tube, by using replica SLFs made from paper towels. As everything worked as planned, we decided to move forward with our design, with only small tolerancing changes.

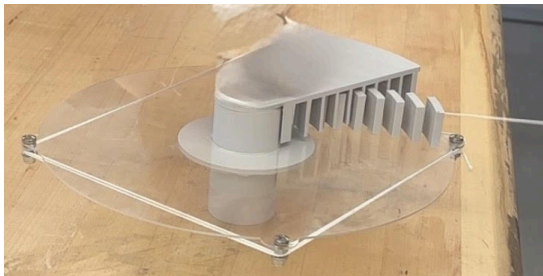


Figure #: String being pulled to spin

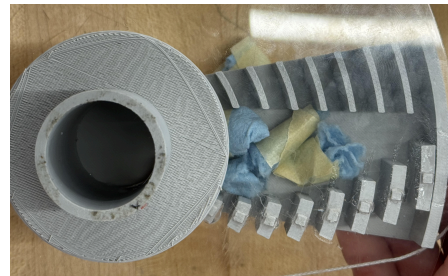


Figure #: Testing threat of death

Component list:

Design Documentation:

Part Name	Specs	McMaster Code	Fabrication
Ridges #1-9	0.875" tall, widths vary from .25"-1.25"	N/A	3D printed out of PLA and done at the RPL
Roof and Wall	r = 4.5", matching ridges 3d printed with roof as one piece	N/A	3D printed out of PLA and done at the RPL
Crown and evoloid gears	96 and 8 teeth	N/A	3D printed out of PLA and done at the RPL
Crank	4" long	N/A	3D printed out of PLA and done at the RPL
Trap Body	9" diameter, 9.5" height	N/A	3D printed out of PLA and done at the RPL
Containment Tub	295.3 in ³ volume	N/A	3D printed out of PLA and done at the RPL
Clear Scratch- and UV-Resistant Cast Acrylic Sheet	12" x 12" x 1/16"	8560K171	Laser cut at RPL
Bolt & Nut	M3 from TDS	N/A	N/A

BOM

Part	Price
Light	\$19.99
3D Prints	\$70.62
Acrylic	\$13.24
Wood	\$1.75
Total	\$105.60

Services

Job	Price
Laser Cutting (for acrylic)	\$31.41